COMP4616 – Fundamentals of Machine Learning

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WEEK 4: BAYESIAN LEARNING

Probability Definitions

- A probability is a measure over a set of events that satisfies three axioms:

1. The measure of each event is between 0 and 1.

   $0 \leq P(X = x_i) \leq 1$, where X is a random variable representing an event and x_i are the possible values of X.

   In general, random variables are denoted by uppercase letters and their values by lowercase letters.

2. The measure of the whole set is 1.

$$\sum_{i=1}^{n} P(X = x_i) = 1$$

3. The probability of a union of disjoint events is the sum of the probabilities of the individual events.

   $P(X = x_1 \vee X = x_2) = P(X = x_1) + P(X = x_2)$, in the case where x_1 and x_2 are disjoint.

Probability Definitions - Probability distribution

- A distribution is a **TABLE of probabilities of values**
- A probability is just a single number
  - P(W=fog) = 0.3

$$\forall x \;\; P(X = x) \geq 0$$

$$\sum_x P(X = x) = 1$$

$P(T)$

T	P
hot	0.5
cold	0.5

$P(W)$

W	P
sun	0.6
rain	0.1
fog	0.3
meteor	0.0

Probability Definitions – Joint distribution

- We need notation for distributions on multiple variables.
- Commas are used for this.
- For example, P(Temperature, Weather) denotes the probabilities of all combinations of the values of Temperature and Weather.
- A joint distribution over a set of random variables: specifies a real number for each assignment (or outcome):

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(X_1 = x_1, X_2 = x_2, \ldots X_n = x_n)$$
$$P(x_1, x_2, \ldots x_n)$$

Must obey the rules

$$P(x_1, x_2, \ldots x_n) \geq 0$$
$$\sum_{(x_1, x_2, \ldots x_n)} P(x_1, x_2, \ldots x_n) = 1$$

Probability Definitions - Events

- An event is a set **E** of outcomes

$$P(E) = \sum_{(x_1 \ldots x_n) \in E} P(x_1 \ldots x_n)$$

$$P(T, W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

- From a joint distribution, we can calculate the probability of any event
  - Probability that it's hot AND sunny? 0.4
  - Probability that it's hot? 0.4 + 0.1 = 0.5
  - Probability that it's hot OR sunny? P(T=hot) + P(W=sun) - P(T=hot, W=sun) = 0.5 + 0.6 - 0.4 =0.7

Probability Definitions - Events

- P(+x, +y)? 0.2
- P(+x)? 0.2+0.3 =0.5
- P(-y OR +x)? P(-y) + P(+x) - P(-y, +x) = 0.4 + 0.5 - 0.3 =0.7

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

Probability Definitions – Marginal distributions

- Marginal distributions are subtables which eliminate variables
- **Marginalization (summing out):** Combine collapsed rows by adding

$$P(T,W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(X_1 = x_1) = \sum_{x_2} P(X_1 = x_1, X_2 = x_2)$$

$$P(t) = \sum_{w} P(t, w)$$

$$P(T)$$

T	P
hot	0.5
cold	0.5

$$P(w) = \sum_{t} P(t, w)$$

$$P(W)$$

W	P
sun	0.6
rain	0.4

Probability Definitions – Marginal distributions

$P(X, Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

$$P(x) = \sum_y P(x, y)$$

$$P(y) = \sum_x P(x, y)$$

$P(X)$

X	P
+x	0.5
-x	0.5

1

$P(Y)$

Y	P
+y	0.6
-y	0.4

1

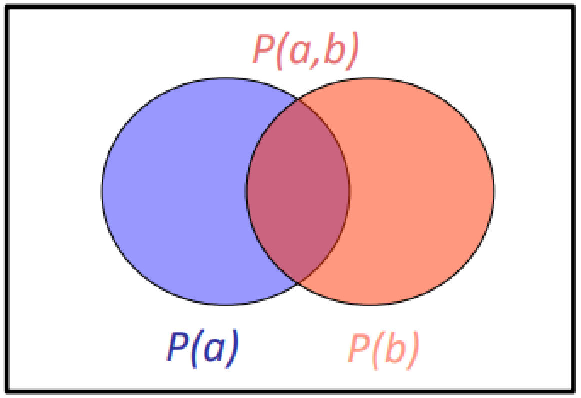
Probability Definitions - Unconditional / Conditional Probabilities

- When rolling **fair dice**, we have P(Total=11) = P((5,6))+P((6,5)) = 1/36+ 1/36 = 1/18.
- Probabilities such as P(Total=11) are called **unconditional** or **prior probabilities** (priors for short).
- **Priors** refer to degrees of belief in proposition in the absence of any other information.
- Most of the time, however, we have some information, usually called **evidence**, that has already been revealed.
- **Example:** We want to roll doubles and we already know that the first die is 5 and the other is spinning at the moment.
  - In that case, we are interested not in the unconditional probability of rolling doubles, but the **conditional** or **posterior probability** (posterior for short) of rolling doubles *given that the first die is a 5*.

Probability Definitions - Conditional Probabilities

- Conditional probabilities are defined in terms of unconditional probabilities as follows:

$$P(a|b) = \frac{P(a,b)}{P(b)}$$



$$P(T,W)$$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W=s|T=c) = \frac{P(W=s,T=c)}{P(T=c)} = \frac{0.2}{0.5} = 0.4$$

$$= P(W=s,T=c) + P(W=r,T=c)$$

$$= 0.2 + 0.3 \quad = 0.5$$

Probability Definitions - Conditional Probabilities

- P(+x | +y)? $P(+x|+y) = \frac{P(+x,+y)}{P(+y)} = \frac{0.2}{0.6} = 0.33$
- P(-x | +y)? $P(-x|+y) = \frac{P(-x,+y)}{P(+y)} = \frac{0.4}{0.6} = 0.66$
- P(-y | +x)? $P(-y|+x) = \frac{P(-y,+x)}{P(+x)} = \frac{0.3}{0.5} = 0.6$

$P(X,Y)$

X	Y	P
+x	+y	0.2
+x	-y	0.3
-x	+y	0.4
-x	-y	0.1

$$P(a|b) = \frac{P(a,b)}{P(b)}$$

Probability Definitions - Conditional Probabilities

- Conditional distributions are probability distributions over some variables given fixed values of others.

Joint Distribution

$P(T, W)$

T	W	P
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(a|b) = \frac{P(a,b)}{P(b)}$$

$P(W|T = hot)$

$$P(W = sun|T = hot) = \frac{0.4}{0.4+0.1} = 0.8$$

$$P(W = rain|T = hot) = \frac{0.1}{0.4+0.1} = 0.2$$

$P(W|T = cold)$

$$P(W = sun|T = cold) = \frac{0.2}{0.2+0.3} = 0.4$$

$$P(W = rain|T = cold) = \frac{0.3}{0.2+0.3} = 0.6$$

Conditional Distributions

$P(W|T)$

$P(W|T = hot)$

W	P
sun	0.8
rain	0.2

$P(W|T = cold)$

W	P
sun	0.4
rain	0.6

$$P(a|b) = \frac{P(a,b)}{P(b)}$$

Probability Definitions - Product Rule

- The definition of conditional probability, can be written in a different form called the **product rule**.

$$P(a,b) = P(a|b)P(b)$$

- By using the product rule, we can obtain the joint distribution from the conditional distribution

Probability Definitions – Chain Rule

- More generally, we can always write any joint distribution as an incremental product of conditional distributions.

$$P(x_1, x_2, x_3) = P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, \ldots x_n) = \prod_i P(x_i|x_1 \ldots x_{i-1})$$

$$P(x_1, x_2) \rightarrow P(x_1)P(x_2|x_1)$$

$$P(x_1, x_2, x_3) \rightarrow P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)$$

$$P(x_1, x_2, x_3, x_4) \rightarrow P(x_1)P(x_2|x_1)P(x_3|x_1, x_2)P(x_4|x_1, x_2, x_3)$$

Bayes Theorem

- In ML, we aim to determine the best hypothesis from some space **H**, given the observed training data **D**.
- One way to specify the best hypothesis is to say that we demand the *most probable hypothesis*, given the data **D** plus any initial knowledge about the ***prior probabilities*** of the various hypotheses in **H**.
- ***Bayes theorem*** provides a way to calculate the probability of a hypothesis based on
  - its prior probability,
  - the probabilities of observing various data given the hypothesis,
  - and the observed data itself

Bayes Theorem - Notations

- **P(h)** denotes the initial probability that hypothesis h holds, before we have observed the training data
- **P(h)** is called the ***prior probability of h*** and may reflect any background knowledge we have about the chance that **h** is a correct hypothesis
- **P(D)** denotes the prior probability that training data **D** will be observed (i.e., the probability of D given no knowledge about which hypothesis holds)
- **P(D | h)** denotes the probability of observing data **D** given some world in which hypothesis **h** holds.
- More generally, we write P(x | y) to denote the probability of x given y.
- In ML problems we are interested in the probability **P (h | D)** that h holds given the observed training data D.
- **P (h | D)** is called the ***posterior probability of h***, because it reflects our confidence that h holds after we have seen the training data D.

Bayes Theorem

- **Bayes theorem** provides a way to calculate the posterior probability **P(h l D)**, from the prior probability **P(h)**, together with **P(D)** and **P(D | h)**

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

- Why we need this?
  - Let us build one conditional from its reverse
  - Often one conditional is tricky but the other one is simple
  - This formula helps in computing the **probability of a cause** given that an **effect has been observed**.

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

Bayes Theorem

This formula helps in computing the probability of a cause given that an effect has been observed.

Medical diagnosis problem

- Example: Diagnostic probability from causal probability:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Example:
  - M: meningitis, S: stiff neck

$$\left.\begin{array}{l} P(+m) = 0.0001 \\ P(+s|+m) = 0.8 \\ P(+s|-m) = 0.01 \end{array}\right\} \text{Example givens}$$

The goal is to compute the probability of having meningitis given a stiff neck.

$$P(+m|+s) = \frac{P(+s|+m)P(+m)}{P(+s)} = \frac{P(+s|+m)P(+m)}{P(+s|+m)P(+m) + P(+s|-m)P(-m)} = \frac{0.8 \times 0.0001}{0.8 \times 0.0001 + 0.01 \times 0.999}$$